

# TMP

记

$$\phi_t = \phi_{\text{FAVOR}+}(k_t), \quad \psi_t = \phi_{\text{FAVOR}+}(q_t), \quad S_t = (W_t, z_t).$$

## 1. 原始Performer

### Update

$$\text{Update}_{\text{perf}}(S_{t-1}, k_t, v_t) : \begin{cases} W_t = W_{t-1} + v_t \phi_t^\top, \\ z_t = z_{t-1} + \phi_t. \end{cases}$$

### Output

$$\text{Output}_{\text{perf}}(S_t, q_t) = \frac{W_t \psi_t}{z_t^\top \psi_t}.$$

## 2. 按 overwrite 思路得到的 Performer + delta rule

### 2.1 先删旧值

设删除后的中间状态为  $S_t^- = (W_t^-, z_t^-)$ ，并假设更新是 rank-1 形式：

$$W_t^- = W_{t-1} + a_t^- \phi_t^\top, \quad z_t^- = z_{t-1}.$$

要求

$$\text{Output}(S_t^-, k_t) = 0.$$

代入：

$$0 = \frac{W_t^- \phi_t}{z_t^{-\top} \phi_t} = \frac{W_{t-1} \phi_t + a_t^- (\phi_t^\top \phi_t)}{z_{t-1}^\top \phi_t}.$$

所以

$$a_t^- = -\frac{W_{t-1} \phi_t}{\|\phi_t\|^2}.$$

于是

$$W_t^- = W_{t-1} - \frac{W_{t-1} \phi_t \phi_t^\top}{\|\phi_t\|^2}, \quad z_t^- = z_{t-1}.$$

### 2.2 再写新值

设最终状态  $S_t = (W_t, z_t)$  满足

$$W_t = W_t^- + a_t^+ \phi_t^\top, \quad z_t = z_t^-.$$

要求

$$\text{Output}(S_t, k_t) = v_t.$$

代入：

$$v_t = \frac{W_t \phi_t}{z_t^\top \phi_t} = \frac{W_t^- \phi_t + a_t^+ (\phi_t^\top \phi_t)}{z_{t-1}^\top \phi_t}.$$

因为

$$W_t^- \phi_t = 0,$$

所以

$$a_t^+ = \frac{z_{t-1}^\top \phi_t}{\|\phi_t\|^2} v_t.$$

因此

$$W_t = W_t^- + \frac{z_{t-1}^\top \phi_t}{\|\phi_t\|^2} v_t \phi_t^\top, \quad z_t = z_{t-1}.$$

合并后得到：

**Update**

$$\text{Update}_\Delta(S_{t-1}, k_t, v_t) : \begin{cases} W_t = W_{t-1} - \frac{W_{t-1} \phi_t \phi_t^\top}{\|\phi_t\|^2} + \frac{z_{t-1}^\top \phi_t}{\|\phi_t\|^2} v_t \phi_t^\top, \\ z_t = z_{t-1}. \end{cases}$$

**Output**

$$\text{Output}_\Delta(S_t, q_t) = \frac{W_t \psi_t}{z_t^\top \psi_t}.$$